AH National Qualifications 2025

X835/77/11

Graphic Communication
Supplementary sheets

WEDNESDAY, 30 APRIL

1:00 PM – 3:30 PM

Supplementary sheet 1 for use with question 1, supplementary sheets 2 and 3 for use with question 3 and supplementary sheet 4 for use with question 5.

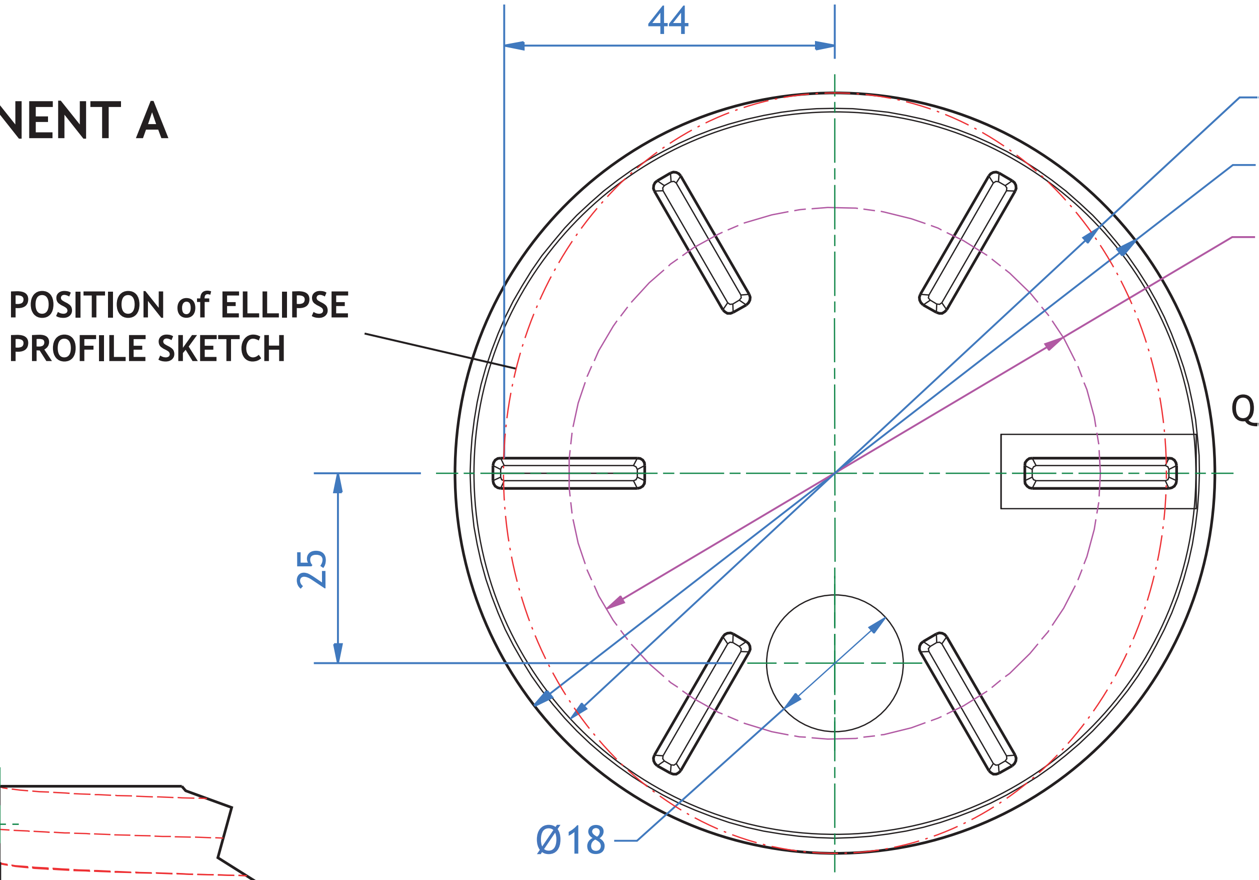
X835771 1

[BLANK PAGE]

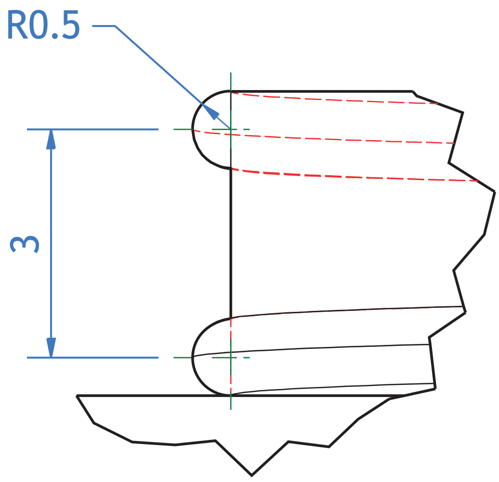
DO NOT WRITE ON THIS PAGE

Supplementary sheet 1 for use with question 1

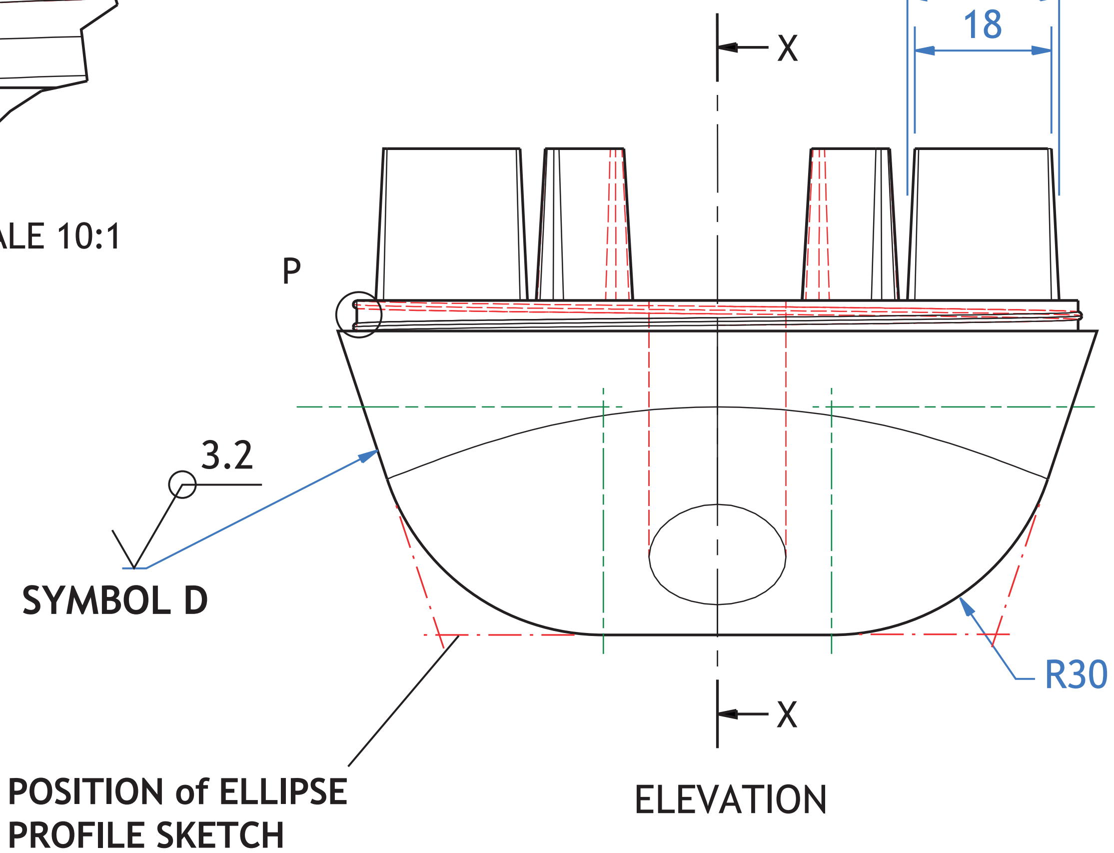
COMPONENT A



PLAN



ENLARGED VIEW P SCALE 10:1



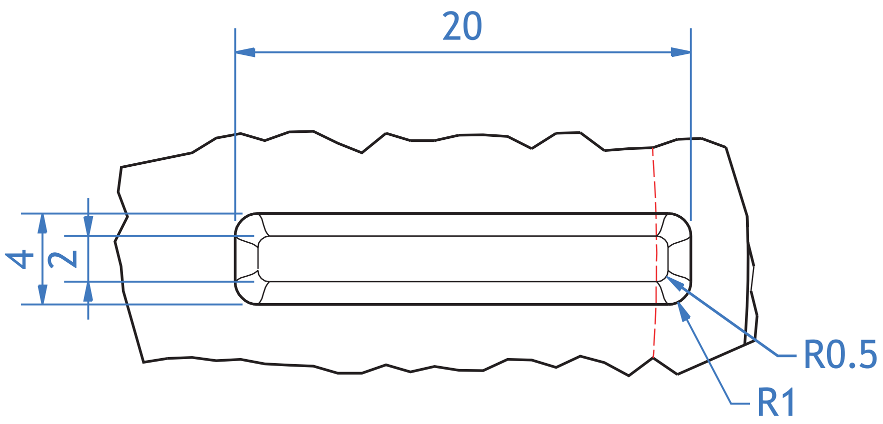
ELEVATION

Ø95 ± 0.15

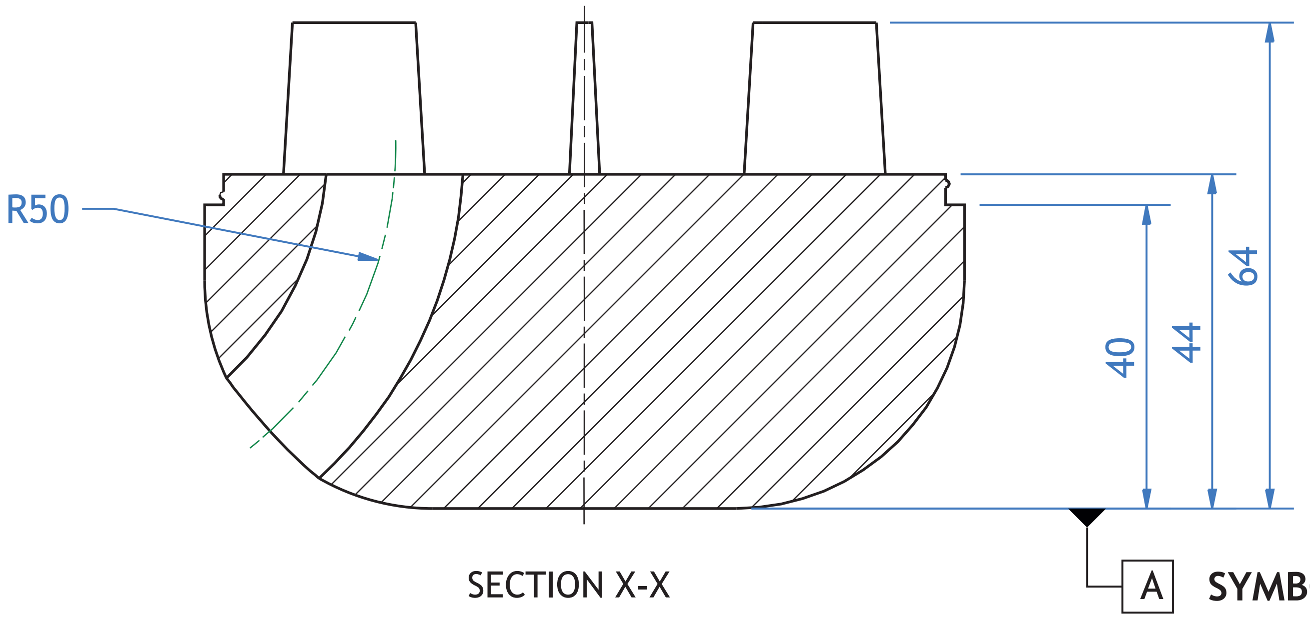
TOLERANCE G

Ø100

PCD 70



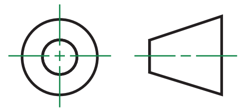
ENLARGED VIEW Q SCALE 3:1 (CENTRE LINES REMOVED FOR CLARITY)



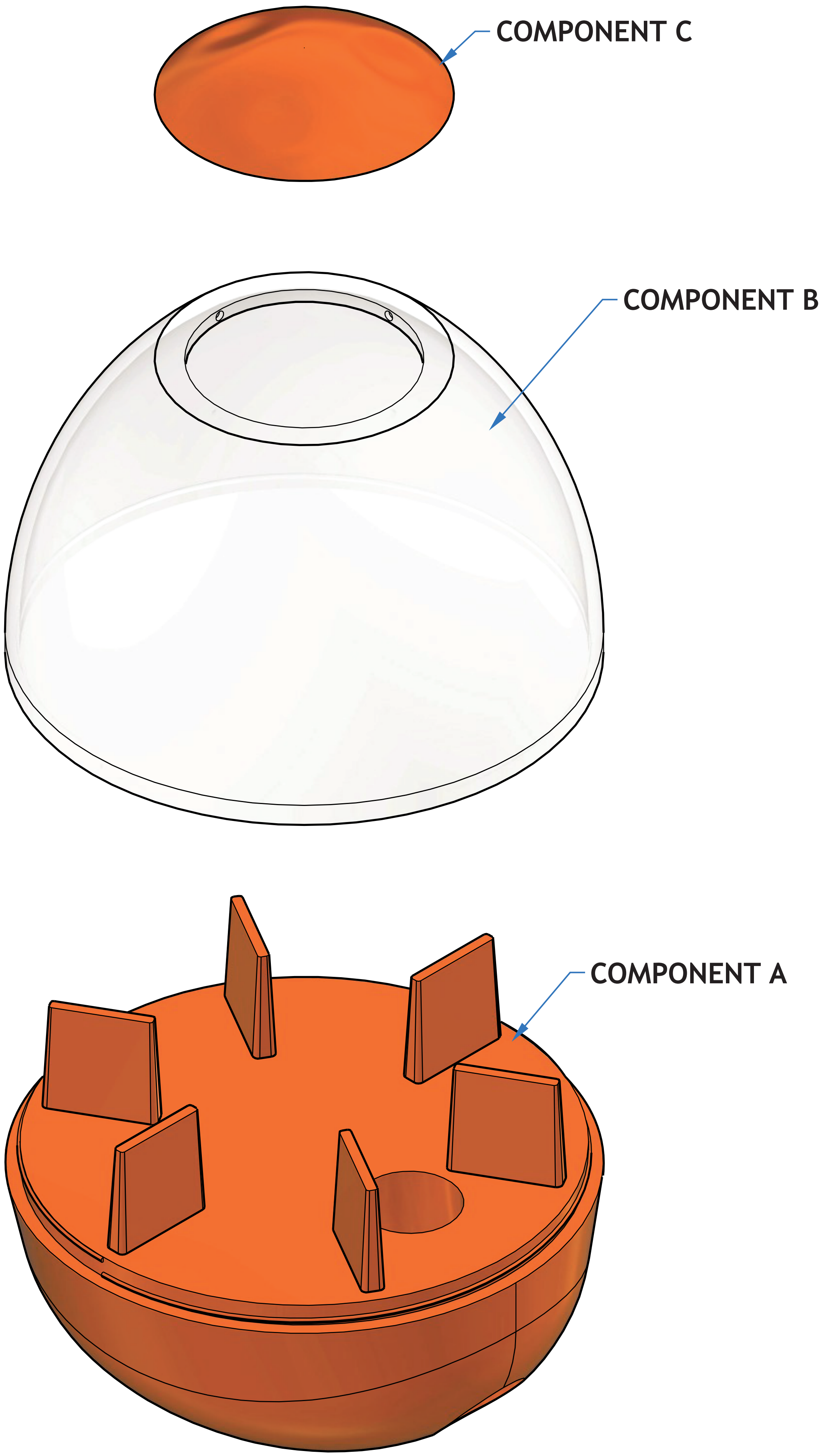
SECTION X-X

SYMBOL E

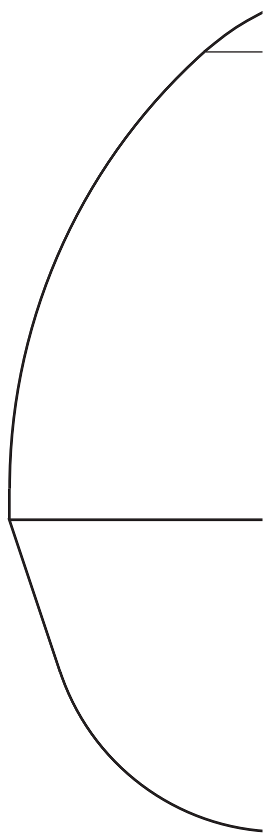
TOLERANCE F

TITLE: DOG TREAT TOY	
DRAWING NUMBER: 1	
ALL DIMENSIONS IN MM TOLERANCE ± 0.2 UNLESS OTHERWISE STATED	SCALE: 1:1
AUTHOR: A.NON	DATE: 10/10/23

Supplementary sheet 3 for use with question 3

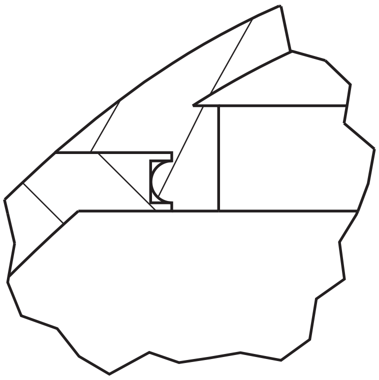


EXPLODED

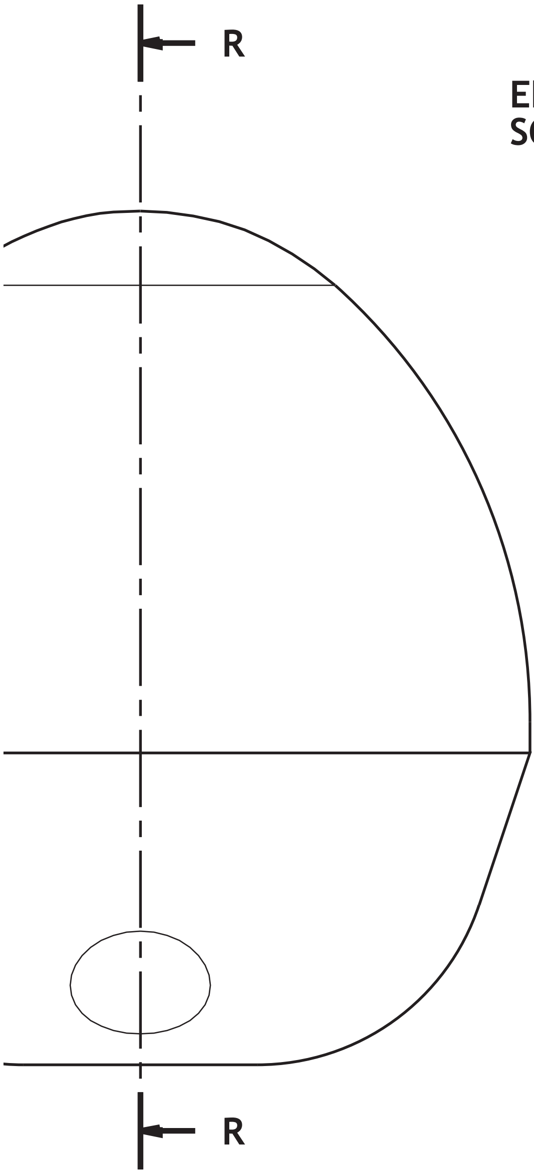


ASSEMBLED

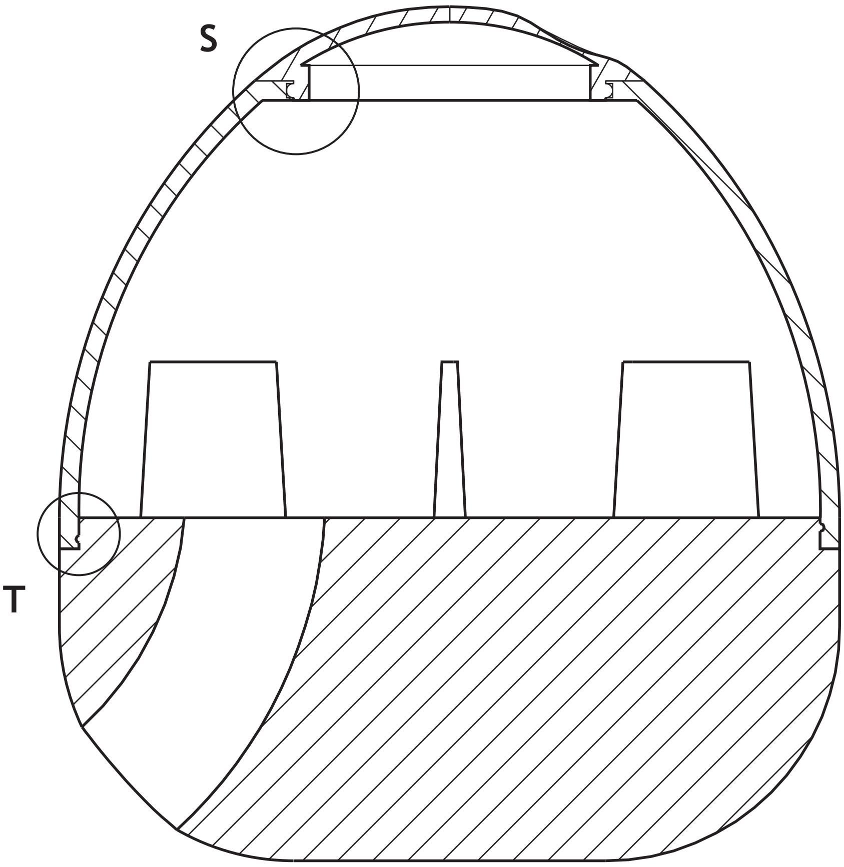
CENTRE LINES HAVE BEEN REMOVED FOR CLARITY



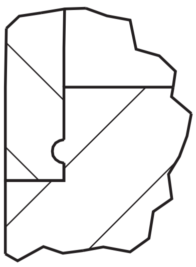
ENLARGED VIEW S
SCALE 4:1



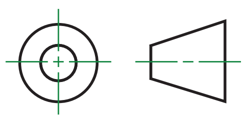
ELEVATION



SECTION R-R



ENLARGED VIEW T
SCALE 4:1

TITLE: DOG TREAT TOY	
DRAWING NUMBER: 2	
EXPLODED AND ORTHOGRAPHIC VIEWS	SCALE: 1:1
AUTHOR: A.NON	DATE: 10/10/23

Supplementary sheet 4 for use with question 5

Specification

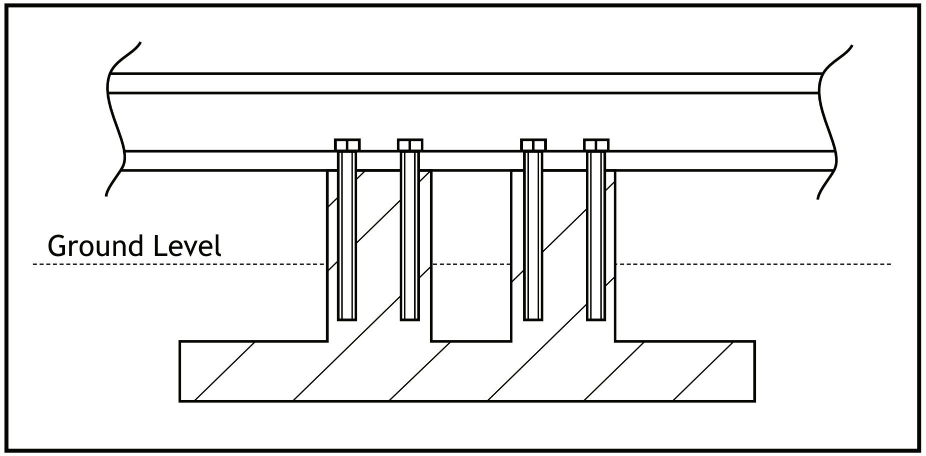
- Footprint: 66×40 m approx
- Height: 20.1 m
- First loop height: 20 m
- Second loop height: 15.4 m
- Track length: 425 m
- Max speed: 18 m/s (65 km/h)
- Acceleration max: 4.2 g
- Run time: 75/80 s
- Number of trains: 1 with 6 cars, 4 seats each
- Track material: Steel
- Foundation material: Concrete
- Foundation connected to supports using fixing bolts
- Frame, supports and tracks assembled using bolts and welding

Joining Methods

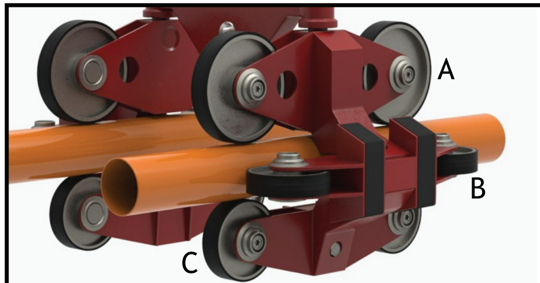
A: Fixing bolts with nut

B: Welded steel supports

Column Foundation (Not to BS8888)



Mechanical wheel and brake system

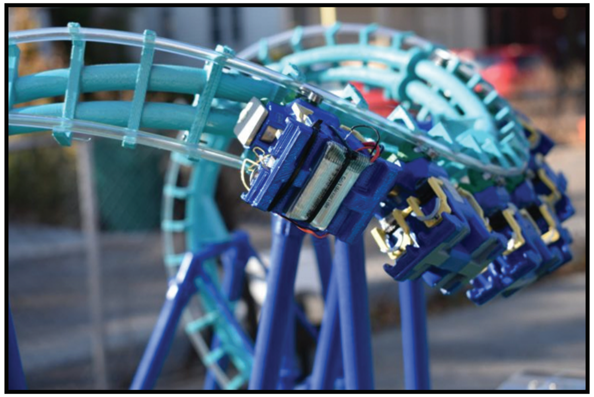


A: Running wheels for guidance

B: Side friction wheels for cornering

C: Up wheels to keep coaster on track

3D Printed model



[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

Acknowledgement of copyright

Supplementary sheet 1	Pay icon: https://pixabay.com/vectors/money-wallet-bank-banknote-bill-3722123/
	Location icon: https://pixabay.com/vectors/location-position-job-you-are-here-1132648/
	Test icon: https://pixabay.com/vectors/checkmark-tick-check-yes-mark-303752/
	Home icon: https://pixabay.com/vectors/house-icon-home-symbol-sign-308936/
	Bike helmet: https://pixabay.com/photos/bicycle-helmet-helmet-bike-helmet-505399/
Supplementary sheet 4	Joining methods: Image is taken from Davide Mavillonio on LinkedIn, https://www.linkedin.com/posts/davide-mavillonio_fem-cfd-ansys-ugcPost-7082302878328995840-Wn4t/
	SQA has made every effort to trace the owners of copyright of this item and seek permissions. We are happy to discuss permission requirements and incorporate any missing acknowledgement. Please contact question.papers@sqa.org.uk.
	Column foundation photo: Image is taken from https://clarkreder.com/entertainment/projects/amusement/mystic-timbers-roller-coaster-foundations
	SQA has made every effort to trace the owners of copyright of this item and seek permissions. We are happy to discuss permission requirements and incorporate any missing acknowledgement. Please contact question.papers@sqa.org.uk.
	Mechanical wheel and brake system: Image is taken from https://free3d.com/3d-model/roller-coasterwheel-884.html
	SQA has made every effort to trace the owners of copyright of this item and seek permissions. We are happy to discuss permission requirements and incorporate any missing acknowledgement. Please contact question.papers@sqa.org.uk.
	3D printed model: Image is taken from 3D Printed Working Roller Coaster Model | MatterHackers
	SQA has made every effort to trace the owners of copyright of this item and seek permissions. We are happy to discuss permission requirements and incorporate any missing acknowledgement. Please contact question.papers@sqa.org.uk.
	Rollercoaster in motion model: Image is taken from See a Functional 3D-Printed Roller Coaster at SOLIDWORKS World 2019 | MySolidWorks
	SQA has made every effort to trace the owners of copyright of this item and seek permissions. We are happy to discuss permission requirements and incorporate any missing acknowledgement. Please contact question.papers@sqa.org.uk.